

ECE 124 – Digital Circuits and Systems
Dept. of ECE, Univ. of Waterloo

Lecture Slides: **Set 5 - Part 1**

Contents:

- Terminologies for sequential circuits
- Storage elements

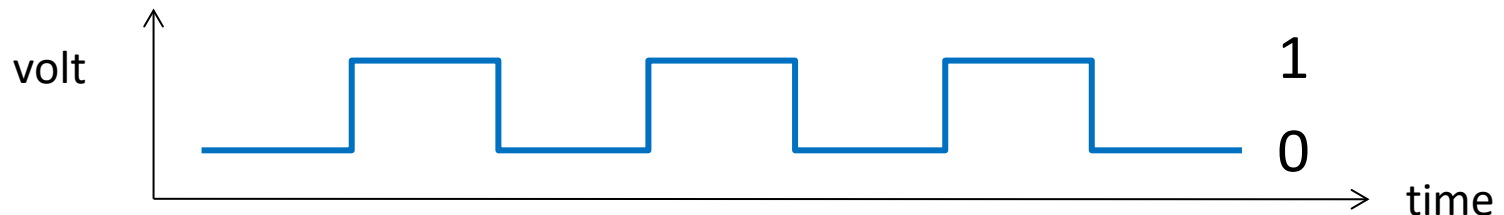
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Introduction

- For the circuits we have seen so far, outputs depend **solely** on present inputs (not on any past signals). Such circuits are known as **combinational** circuits.
- Circuits for which the outputs depend not only on present inputs but also on **past** signals are known as **sequential circuits**.
- Sequential circuits have ‘memory’ or storage elements that can store logic values.
- The content of the storage elements defines the **state** of the sequential circuit.

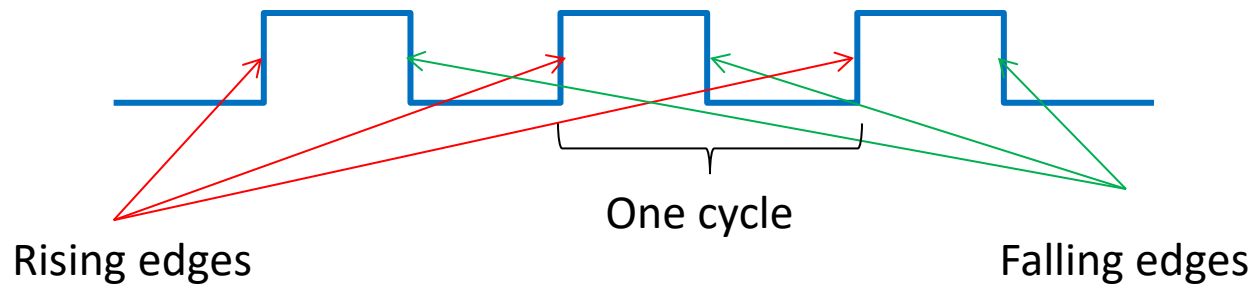
Introduction (contd.)

- There are two main types of sequential circuits
 - *Synchronous* sequential circuits: in this type, the circuit behavior (i.e., signals) changes only at *discrete instances* in time
 - *Asynchronous* sequential circuits: in this type, the circuit behavior can change at *any instance* in time
- A '*clock*' signal is used to change the circuit behavior at discrete instances in time
 - A clock signal has two levels (corresponding to logic 1 and 0) as shown below



Introduction (contd.)

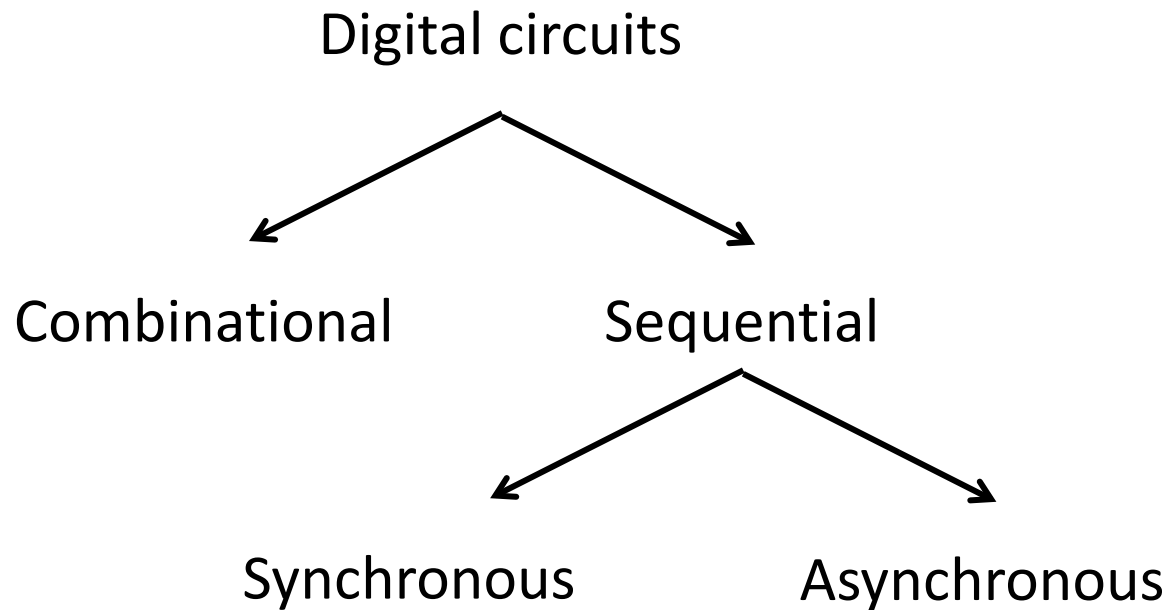
- A $0 \rightarrow 1$ transition in the clock signal is called **rising (positive)** edge
- A $1 \rightarrow 0$ transition in the clock signal is called **falling (negative)** edge



- Time duration (say t) of one cycle is known as clock **period**
- Clock **frequency** (say f) is the reciprocal of t , i.e., $f = 1/t$
- units of t : sec (or micro-, nano-sec, etc.)
- Units of f : Hz (or MHz, GHz, etc.)

Introduction (contd.)

Putting it together: types of digital circuits



Storage elements

There are mainly **two** types of storage elements in sequential circuits: **latches** and **flip-flops**.

Basic latch: A basic latch responds with changes in the signal **level** of input data (i.e., a basic latch is level sensitive)

Gated latch: It is a basic latch with an enable type control input (e.g., clock).

- If the control is not asserted, the gated latch does not respond to changes in the signal level of input data
- On the other hand, if the control is asserted, a gated latch acts like a **level** sensitive basic latch described above
 - For a periodic control signal like a clock, the state can change **multiple** times during a single cycle to respond to changes in the signal level of input data

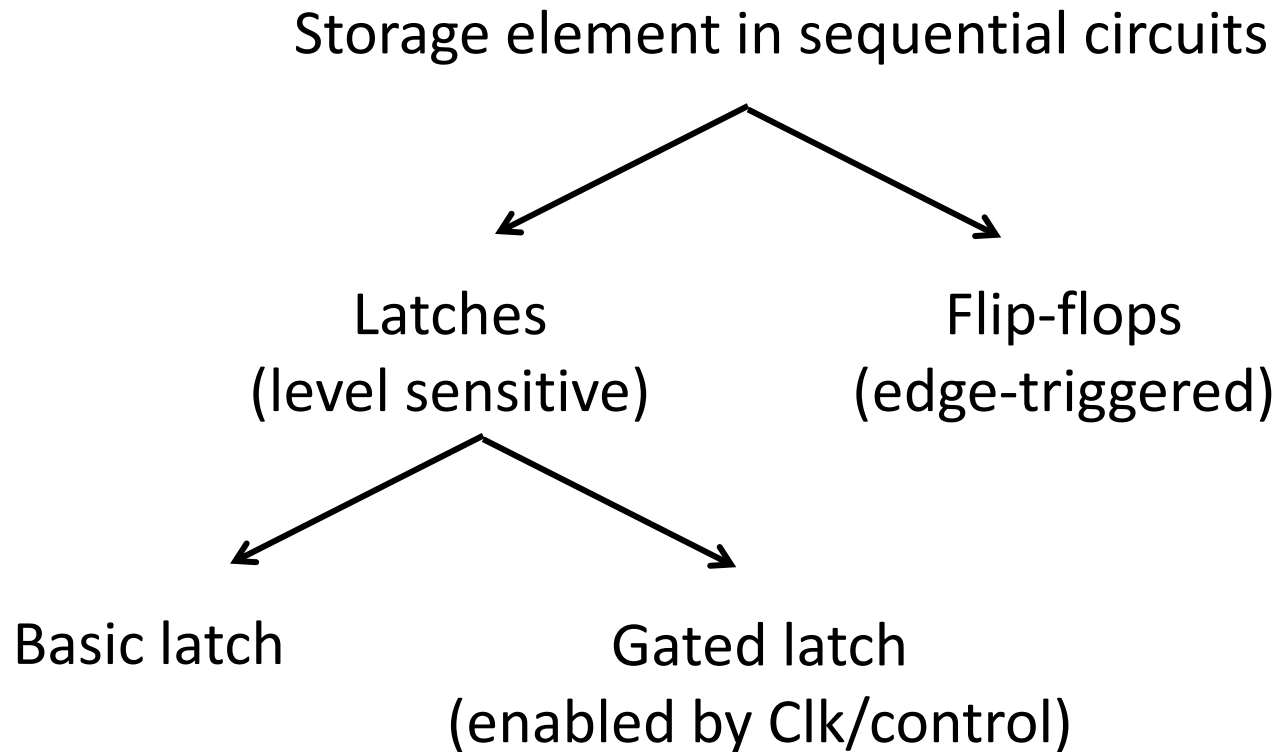
Storage elements (contd.)

Flip-flop:

- A flip-flop has an enable type control input (e.g., clock) and can be viewed to have two gated latches.
- The state of a flip-flop responds to input data signals only at one of the **edges** (rising or falling) of the control signal.
- Since (clock) edges are almost instant in time, a flip-flop can't change its state more than once in a single clock cycle.

Storage elements (contd.)

Putting it together: Storage elements in sequential circuits



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Lecture Slides: **Set 5 - Part 2**

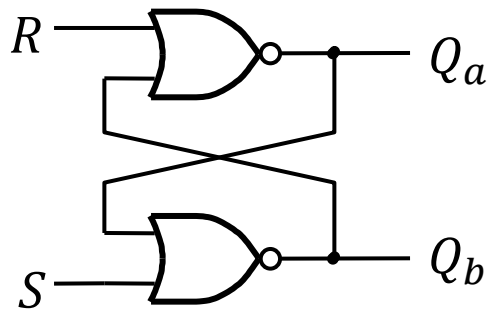
Contents:

- Basic latches
- Gated latches

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Basic latch with NOR gates

- The following circuit made of two NOR gates connected in a crossed-coupled way is a latch (storage element)
- Its behavior is described in the following table
 - for sequential circuits, the table describing the circuit behavior is called characteristic table
 - Inputs are R and S (reset and set respectively)
 - When $(R, S) = (1, 0)$, output Q_a is reset to 0
 - When $(R, S) = (0, 1)$, output Q_a is set to 1
 - Input $(R, S) = (1, 1)$ is avoided (see also the next slide), in which case $Q_a = Q'_b$ and the value of output Q_a is treated as the state of the latch



Circuit

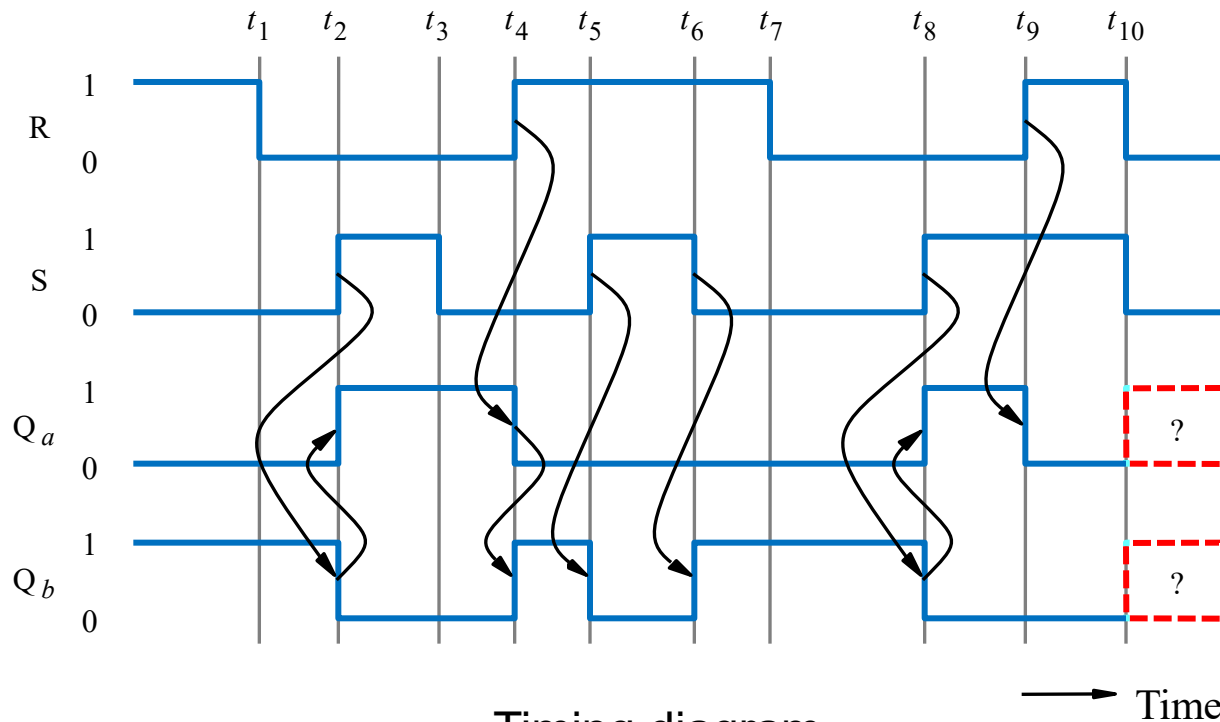
S	R	Q_a	Q_b
0	0	nc	nc
0	1	0	1
1	0	1	0
1	1	0	0

(nc=no change)

Characteristic table

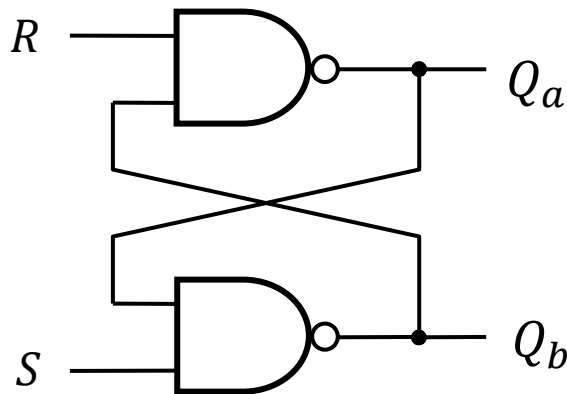
Basic latch with NOR gates (contd.)

- The timing diagram of the NOR based latch is shown below
- When (R, S) changes from $(1, 1)$ to $(0, 0)$, the state will keep toggling (oscillate) between 0 & 1
 - in practice the state will settle into 0 or 1, but we don't know what it will be.



Basic latch with NAND gates

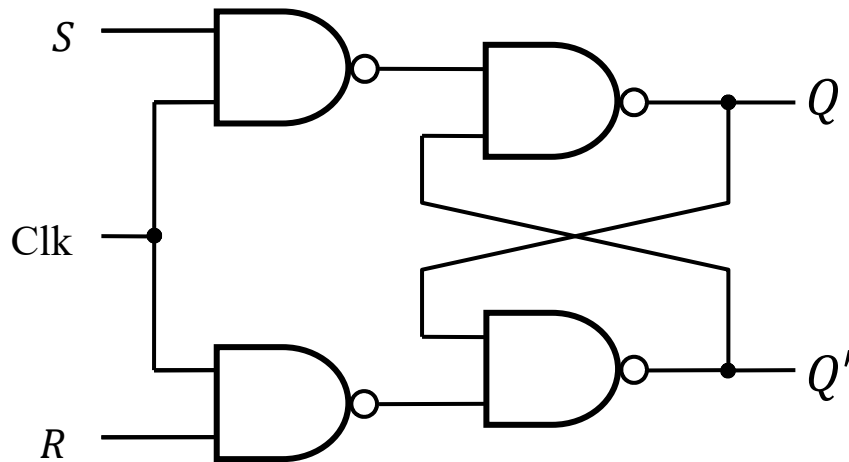
- The following circuit made of two NAND gates connected in a crossed-coupled way is a latch
- Its behavior is described in the following table
 - When (R, S) changes from $(0,0)$ to $(1,1)$, the state will keep toggling (oscillate) between 0 & 1



S	R	Q_a	Q_b
1	1	nc	nc
0	1	0	1
1	0	1	0
0	0	1	1

Gated latch with NAND gates

- In the basic SR latches presented earlier, the state changes can occur anytime and we may not be aware of such changes
- We can control the timing of possible state changes via an Enable or a clock signal
- The following circuit diagram is a **gated** latch.
 - When $\text{Clk}=0$, the gated latch incurs no state change even with changes in S and R
 - When $\text{Clk}=1$, the gated latch functions as a basic latch



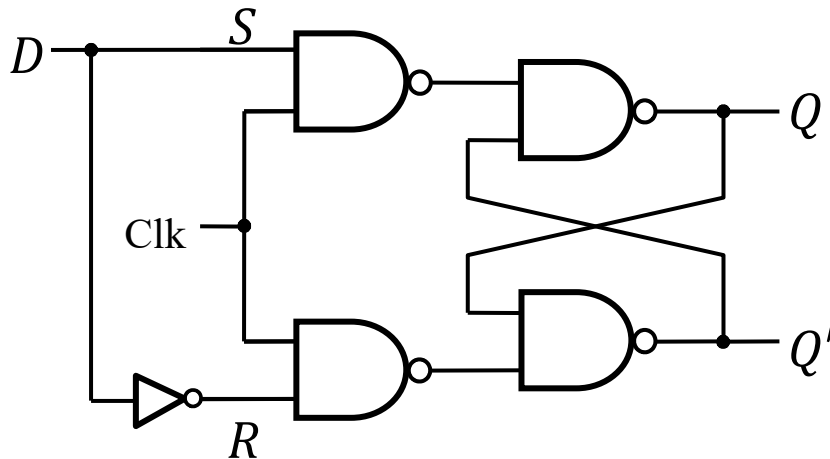
Gated SR latch with NAND gates

Clk	S	R	Q	Q'
0	x	x	nc	nc
1	0	0	nc	nc
1	0	1	0	1
1	1	0	1	0
1	1	1	x	x

Characteristic table

Gated D latch

- Gated D latch is a special case of the gated latch shown in the previous slide and has a **single** data input (D)
- The special case is due to $R = S'$; see the diagram and the characteristic table below, where $Q(t)$ and $Q(t+1)$ denote the present and the next states, respectively
- Gated D latch stores the value of the input data under the control of Clk



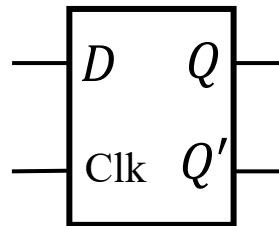
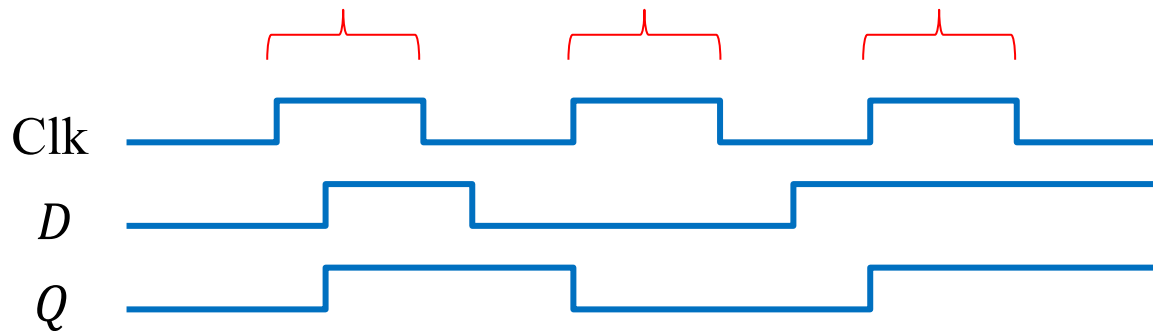
Gated D latch circuit

Clk	D	$Q(t+1)$
0	x	$Q(t)$ (i.e., nc)
1	1	1
1	0	0

Characteristic table

Gated D latch (contd.)

- Gated D latch shown in the previous slide is controlled by the level of Clk (and hence called '*level sensitive*'); see the timing diagram below where the sensitive portions are identified with red curly brackets.
- Graphical symbol the level sensitive D latch is also shown below.



Graphical symbol

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Lecture Slides: **Set 5 - Part 3**

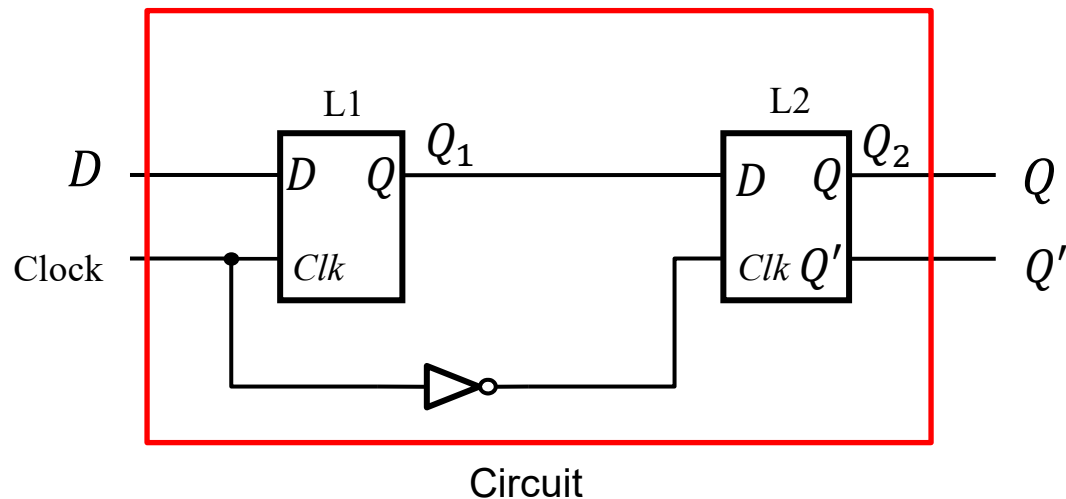
Contents:

- Flip-flops

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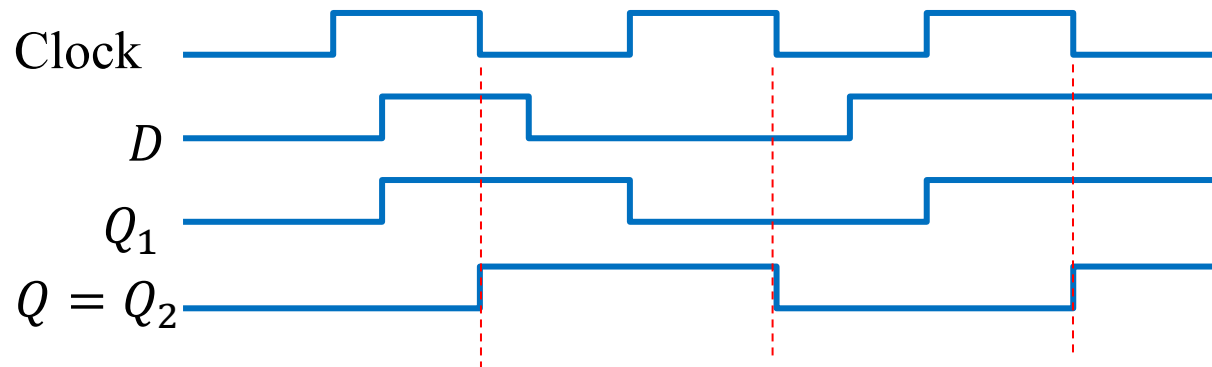
Edge-Triggered Flip-flops

- In the following circuit,
 - the 1st D latch (L1) changes its state while Clock=1.
 - the 2nd D latch (L2) changes its state while Clock=0.
- Hence, while Clock=1, Q_1 follows changes in input data D , and Q_2 remains constant
- When Clock becomes 0, L1 stops following changes in data D and at the same time L2 responds to changes in Q_1



Edge-Triggered Flip-flops (contd.)

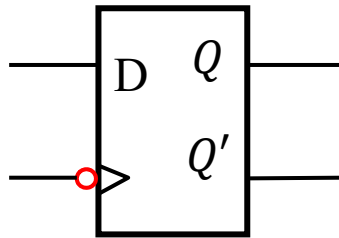
- Since Q_1 does not change while Clock=0, latch L2 can undergo only one change during one clock cycle
- From the external observer's point of view (i.e., outside the red box in the previous slide), the combined circuit changes its state at the negative edge of Clock
- The timing diagram is shown below



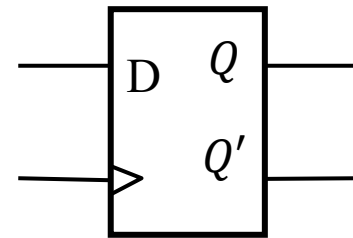
Timing diagram

Edge-Triggered Flip-flops (contd.)

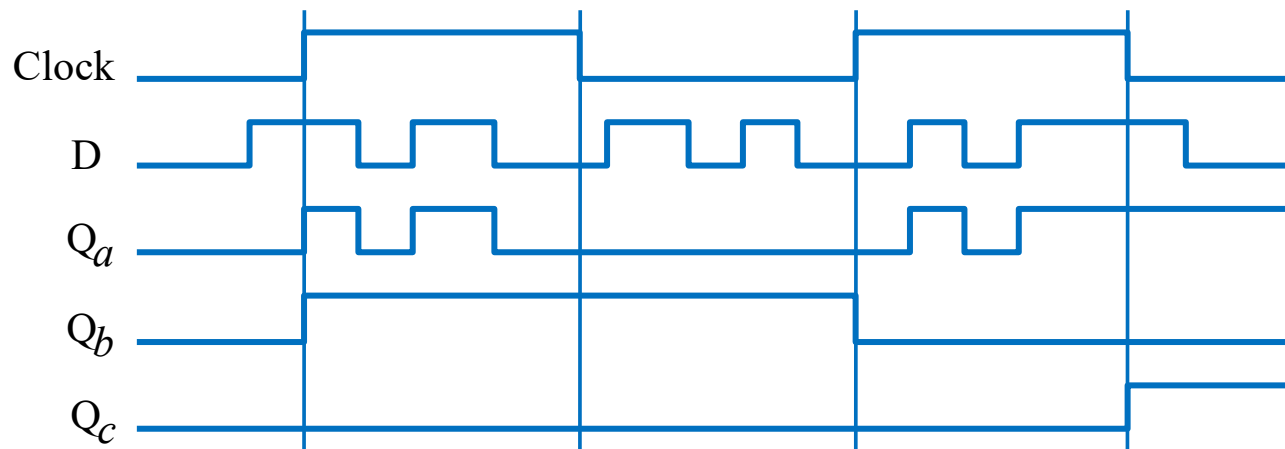
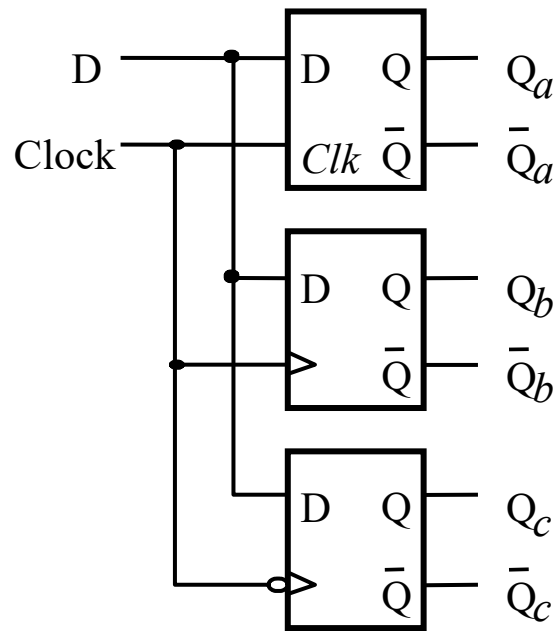
- The previous circuit is a negative-edge-triggered D flip-flop
- The circuit can be modified to a positive-edge-triggered D flip-flop by connecting Clk of latch L2 directly to Clock and Clk of latch L1 to the complement of Clock
- Graphical symbols for the both negative- and positive-edge-triggered D flip-flops are shown below
- Timing diagrams for level-sensitive and edge-triggered D storage elements are shown in the next slide



Negative-edge-triggered
DFF

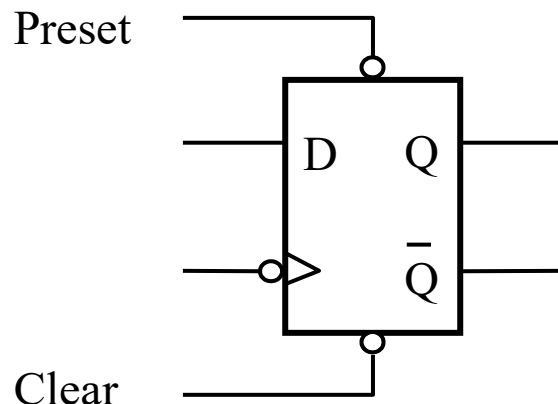


Positive-edge-triggered
DFF



DFF with Clear and Preset

- The circuit for edge-triggered DFF can be modified so that the state of DFF can be **cleared to 0** or **preset to 1**.
- Clear and Preset are **asynchronous** operations
- Graphical symbol is given below
- A synchronous clear operation can be obtained by using a two input AND gate before D. The inputs to the AND gate will be the data input and Clear'



Preset and Clear are active low, i.e.,

- If Preset=0, then Q=1
- If Clear=0, then Q=0
- Both Preset and Clear should **not** be activated at the same time

Timing analysis with flip-flops

- It takes time for gates to change their output values according to the input values – i.e., there are *propagation delays* due to resistance, capacitance, etc.
- Changes in flip-flop outputs occur at the *active* clock edge.
- There are three timing parameters that are especially important:
 - Setup time (t_{su})
 - Hold time (t_h)
 - Clock-to-output time (t_{cQ})

Timing analysis with flip-flops (contd.)

- The *setup time* of a flip-flop is the amount of time that the data inputs need to be held stable (not changing) **PRIOR** to the arrival of the active clock edge.
- The *hold time* of a flip-flop is the amount of time that the data inputs need to be held stable (not changing) **AFTER** the arrival of the active clock edge.
- The *clock-to-output time* of a flip-flop is the amount of time it takes for the output to become stable (at its new value) **AFTER** the arrival of the active clock edge.

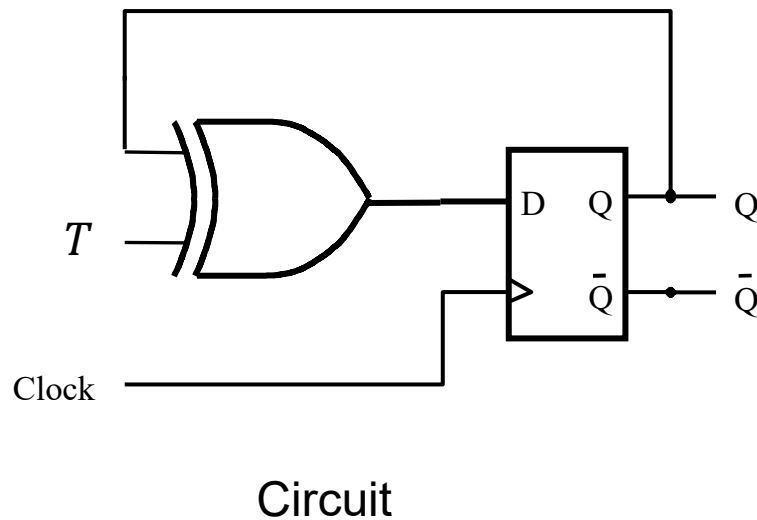
Timing analysis with flip-flops (contd.)

Comments

- If these timing specifications are not met, then it is possible that the flip-flop will not behave as expected.
 - That is, if we don't observe setup and hold times at the data inputs, then our output might not change as expected.
 - That is, if we don't wait long enough (clock-to-output time) for the output to change, then we might use an incorrect value.
- If we fail to meet any of these timing parameters, then we have a **timing violation**.
- These timing parameters have an influence on how fast we can clock a circuit (more to come later)

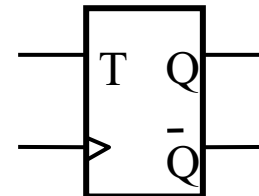
T Flip-flop

- A circuit diagram and the characteristic table along with the graphical symbol of T flip-flop are given below. An example timing diagram is in the next slide
- T flip-flops are useful in some designs



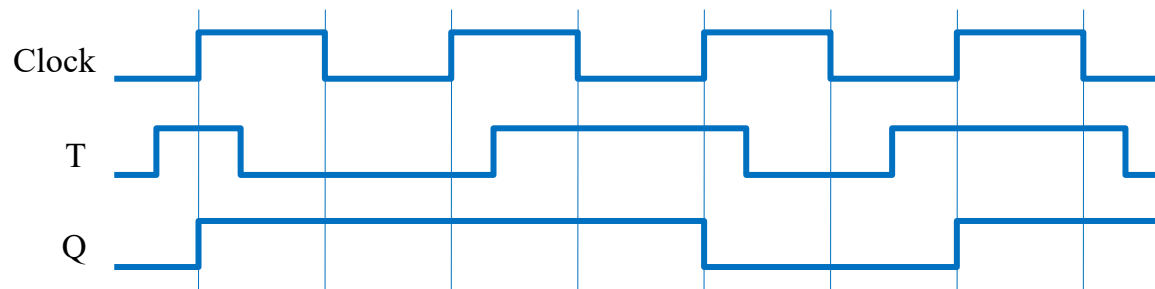
T	$Q(t+1)$
0	$Q(t)$
1	$Q'(t)$

Characteristic table



Graphical symbol

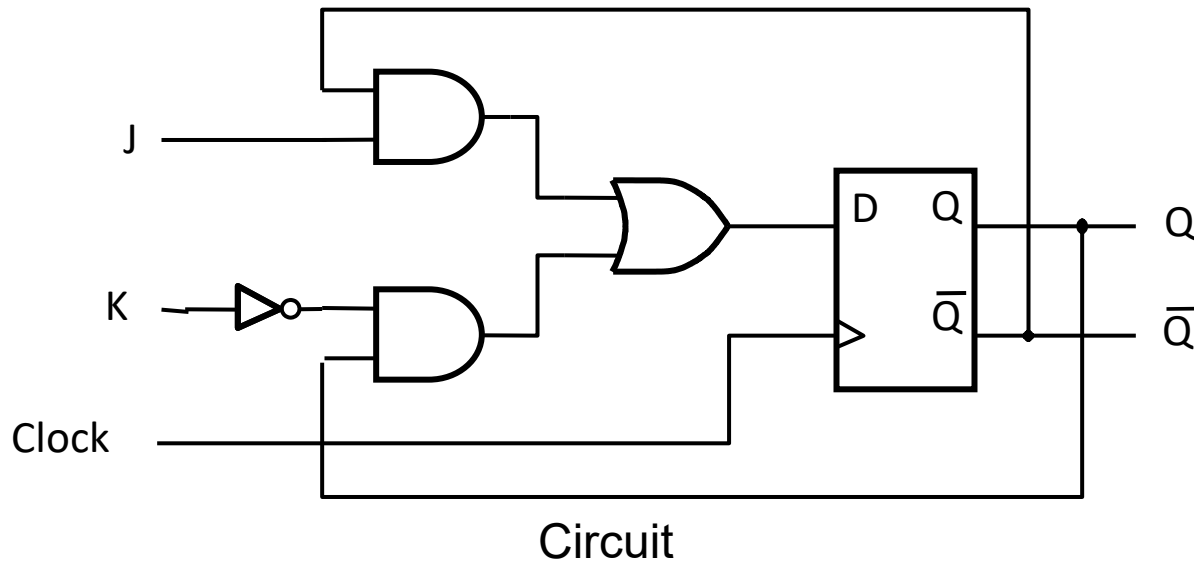
T Flip-flop (contd.)



Timing diagram

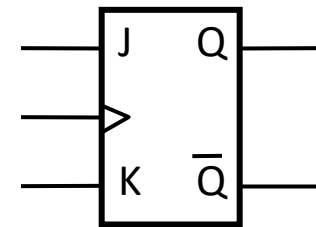
J-K Flip-flop

- J-K flip-flops are generalization of D and T flip-flops. A circuit diagram and the characteristic table along with the graphical symbol of a J-K flip-flop are given below.
- From the table, we see that it's
 - a T flip-flop if $T=J=K$ and
 - A D flip-flop if $D=J=K'$



J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\bar{Q}(t)$

Char. table



Graphical symbol

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Lecture Slides: **Set 5 - Part 4**

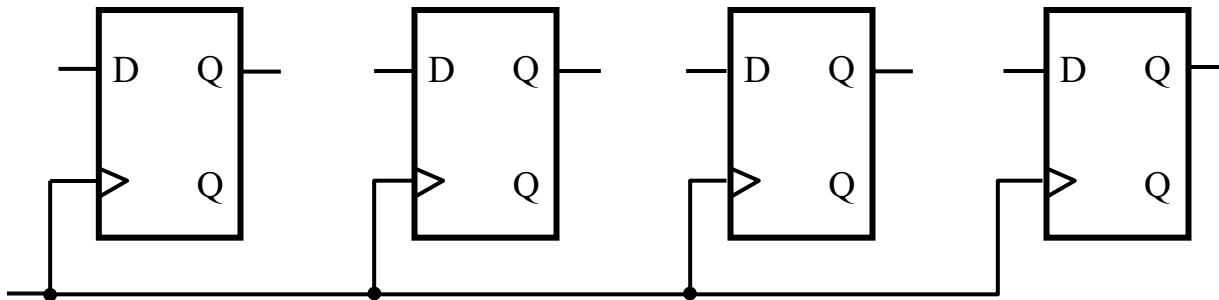
Contents:

- Registers
- Asynchronous counters

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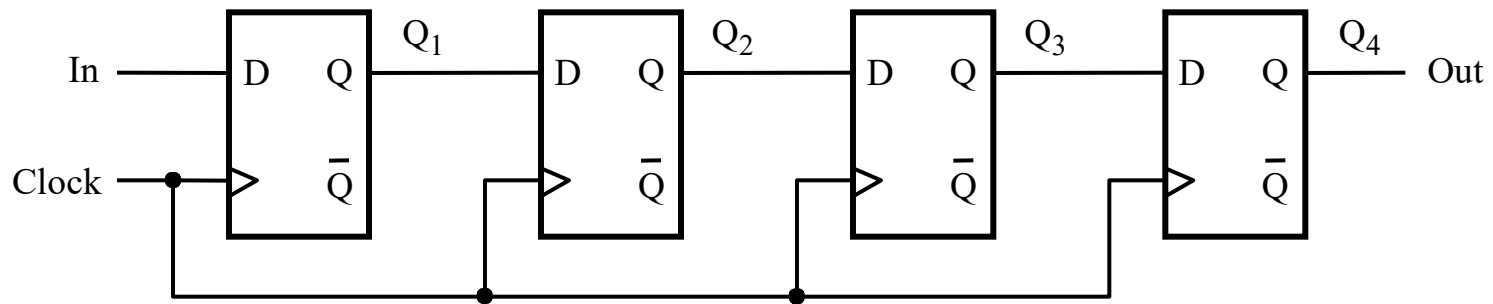
Registers

- A register is a digital storage; an n-bit register stores n bits.
- Registers considered here are made of flip-flops, i.e., a flip-flop is a 1-bit register.
- A 4-bit register made of four D flip-flops is shown below:



- By having additional circuits at the flip-flop inputs and outputs, the register contents can be modified in some special way (e.g., contents can be **shifted, incremented, decremented, etc.**)

Shift registers

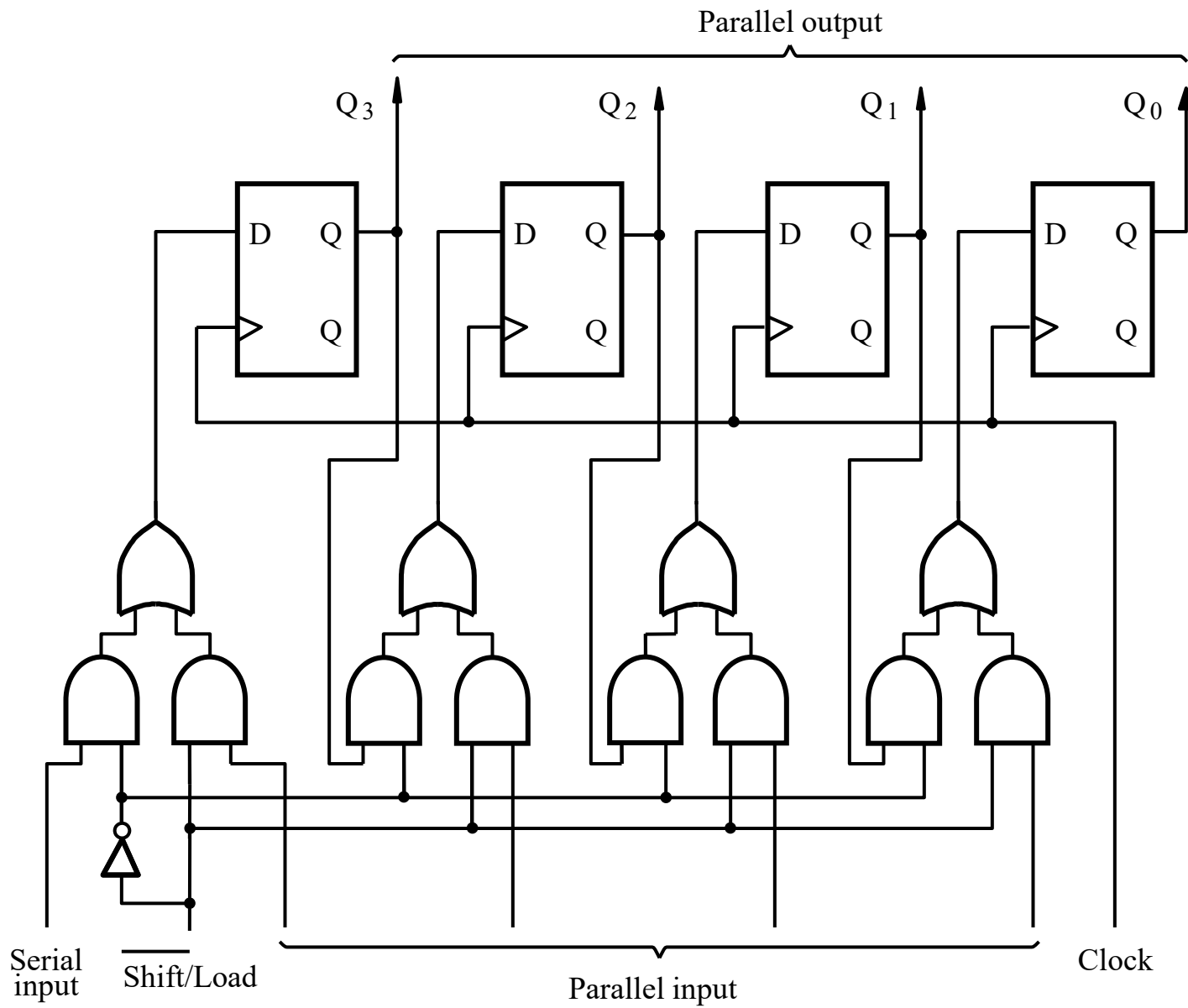


- A basic shift register using D flip-flops is shown above.
- A sample sequence of output is given at right.

	In	Q_1	Q_2	Q_3	$Q_4 = \text{Out}$
t_0	1	0	0	0	0
t_1	0	1	0	0	0
t_2	1	0	1	0	0
t_3	1	1	0	1	0
t_4	1	1	1	0	1
t_5	0	1	1	1	0
t_6	0	0	1	1	1
t_7	0	0	0	1	1

Shift registers (contd.)

- A parallel access shift register using D flip-flops (DFF) is shown in the next slide
- It is very versatile and supports
 - serial as well as *parallel* input (i.e., load)
 - Serial as well as parallel output



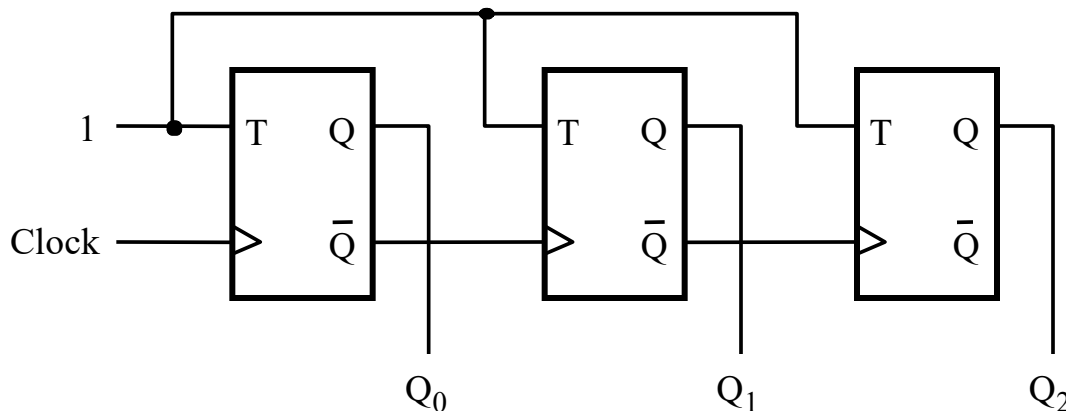
Parallel-access shift register.

Counters

- A counter can increment or decrement (typically by 1) the “number” in its storage
- A counter that increments its number is called an **up**-counter.
- Similarly, a counter that decrements its number is called a **down**-counter.
- For increment/decrement, if the counter updates all the bits of its number at the same time, then it is called a **synchronous** counter; otherwise it's an **asynchronous** counter

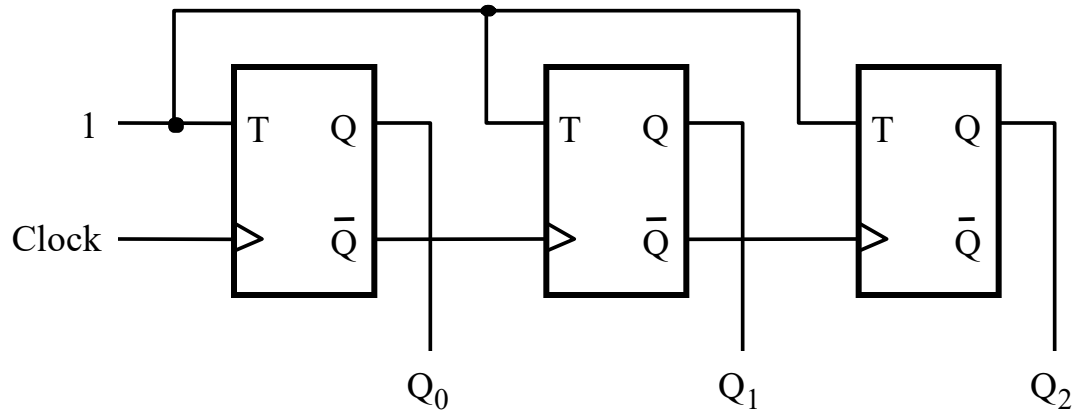
T flip-flop based asynchronous counter

- A three-bit **up**-counter using T flip-flop is shown below.
- The output bits correspond to the binary number $(Q_2Q_1Q_0)_2$, i.e., Q_0 is the least significant bit and Q_2 is the most significant bit
- Note all T inputs are connected to 1
- $\bar{Q} = Q'$ of one flip-flop is the Clk input to the next flip-flop at right
- Its timing diagram is shown in the next slide and there we see that not all output bits change at the same time. Hence it's called an **asynchronous** counter

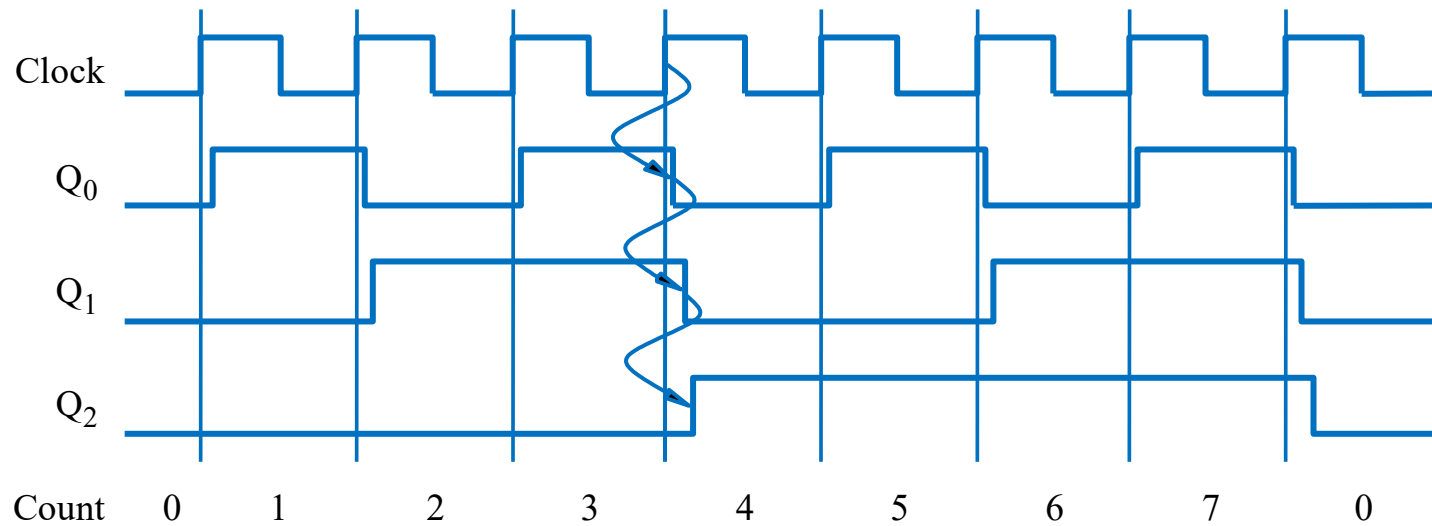


Circuit

Yours to do:
modify the circuit to
make it a **down**-counter



Circuit (from the previous slide)



Timing diagram

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Lecture Slides: **Set 5 - Part 5**

Contents:

- Synchronous counters

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T flip-flop based synchronous counters

Note how an output bit (say Q_2) changes in the table at right

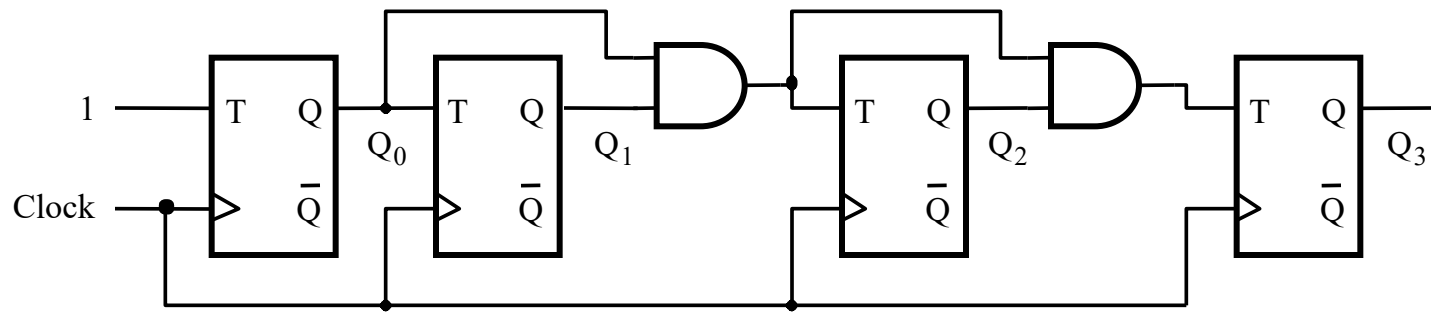
- Q_2 changes (0 to 1, or 1 to 0) when lower significant bits (i.e., Q_1 and Q_0) are both 1
- This can be generalized for other bits

Circuit and timing diagrams of a **synchronous** up-counter using T flip-flops are shown in the next slide

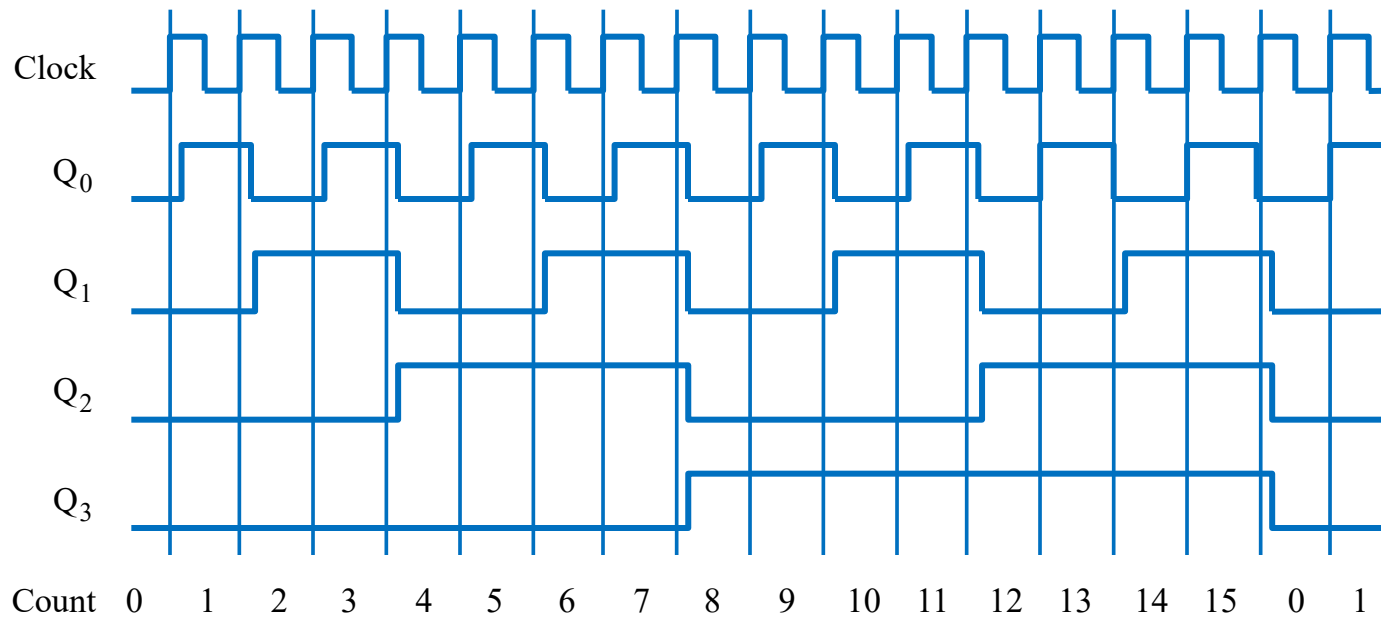
Clock cycle	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Q₁ changes
Q₂ changes

Derivation of a synchronous up-counter



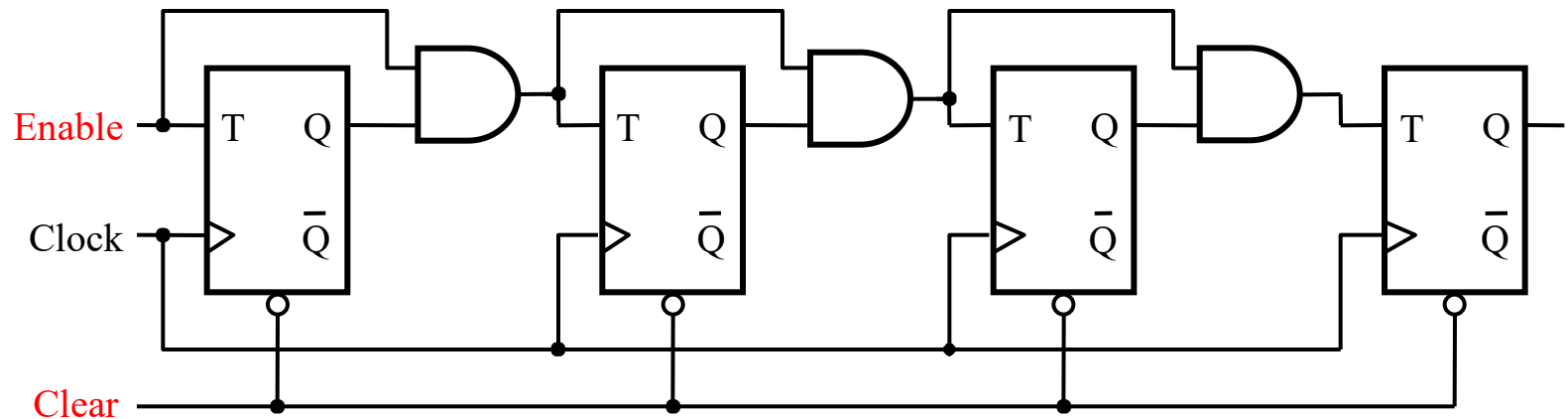
Circuit



Timing diagram

A four-bit synchronous up-counter.

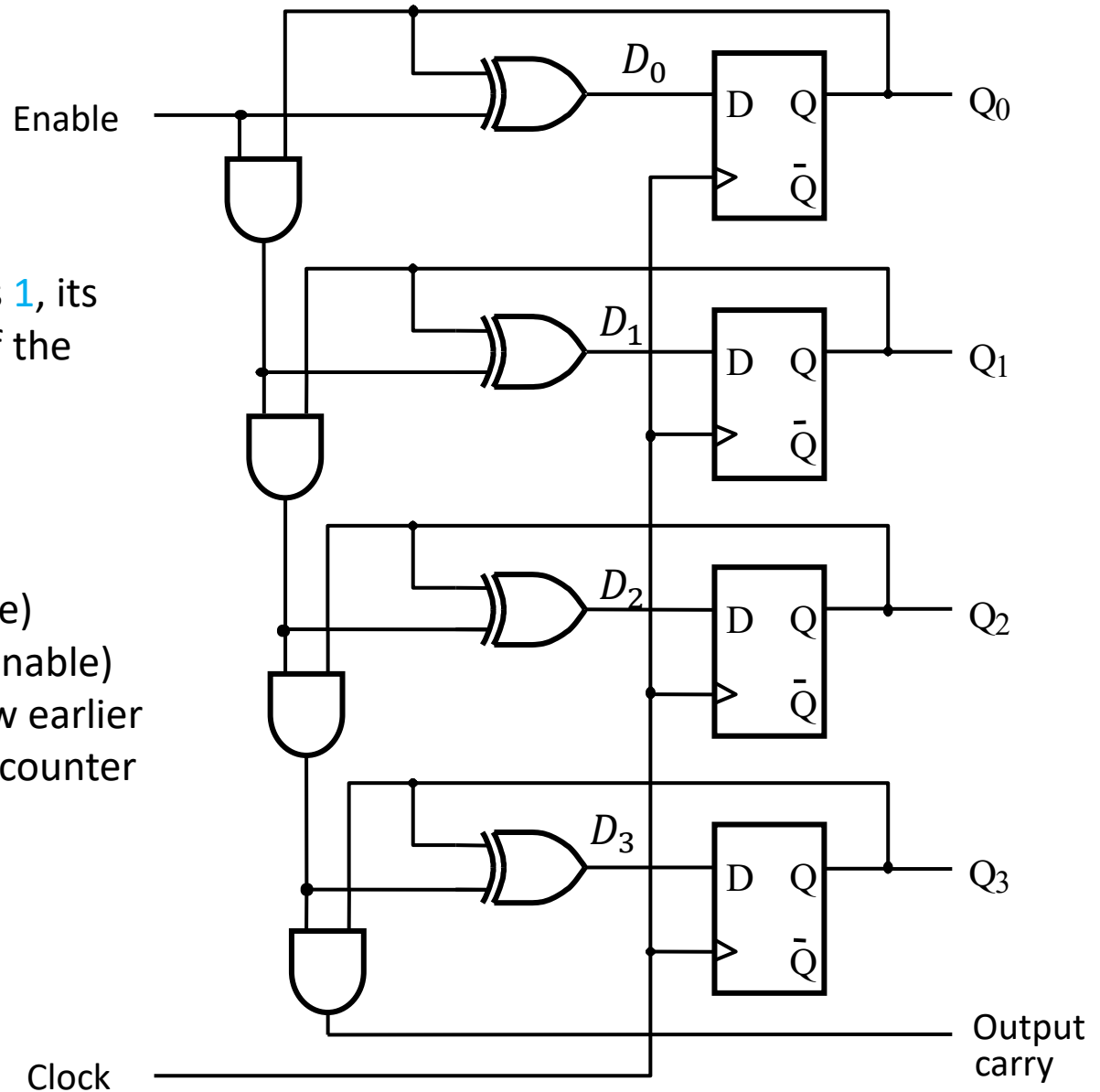
Synchronous counters with Enable and Clear Capability

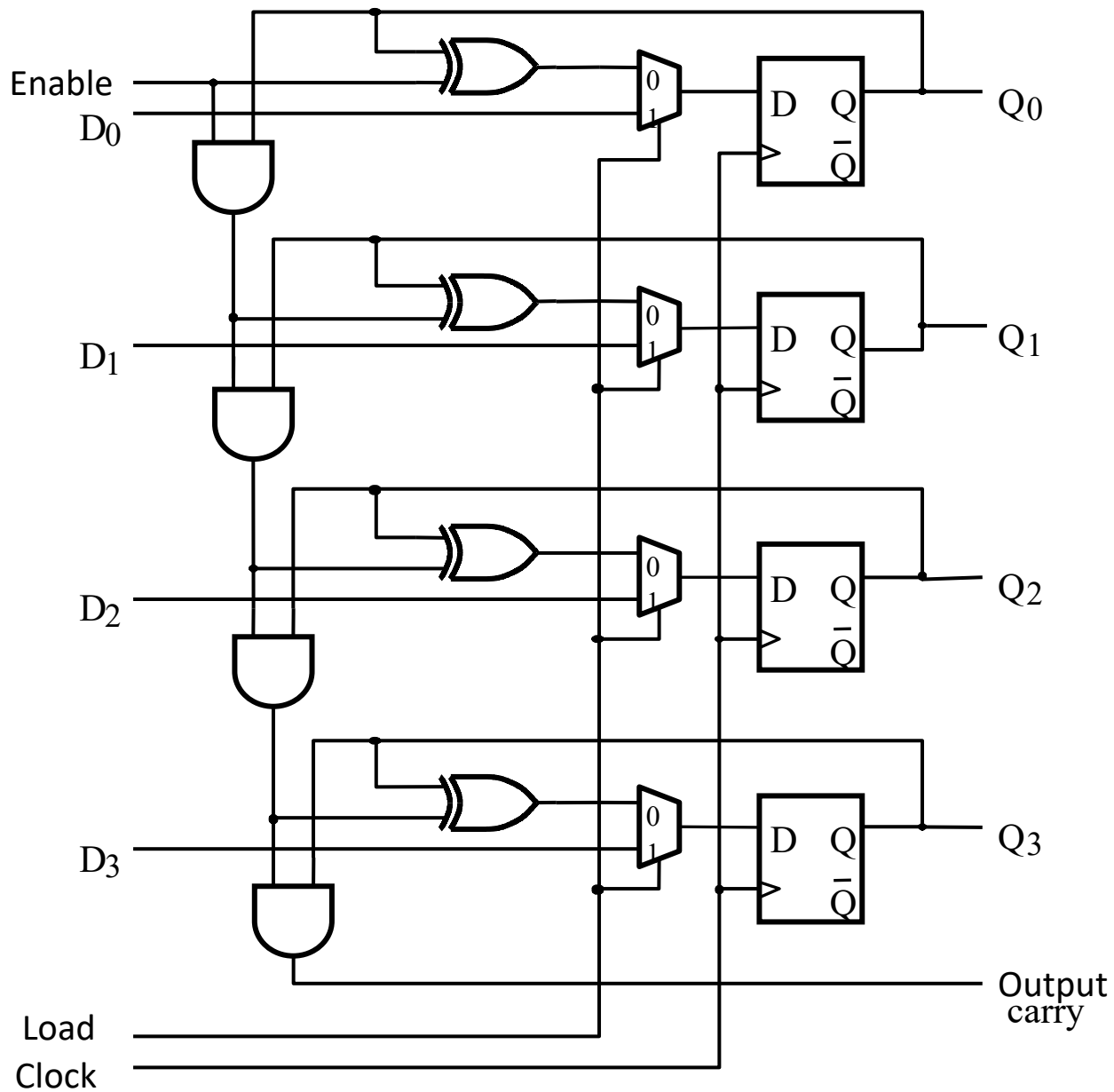


D flip-flop based counters

Note that:

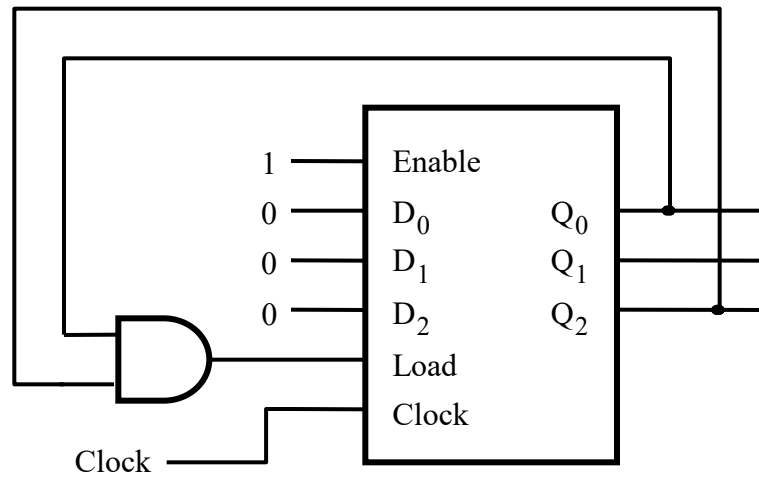
- When one input of an XOR is **1**, its output is the **complement** of the **other input**
- In the diagram
$$D_0 = Q_0 \oplus \text{Enable}$$
$$D_1 = Q_1 \oplus (Q_0 \cdot \text{Enable})$$
$$D_2 = Q_2 \oplus (Q_1 \cdot Q_0 \cdot \text{Enable})$$
$$D_3 = Q_3 \oplus (Q_2 \cdot Q_1 \cdot Q_0 \cdot \text{Enable})$$
- This is similar to what we saw earlier for designing a synchronous counter with T flip-flops



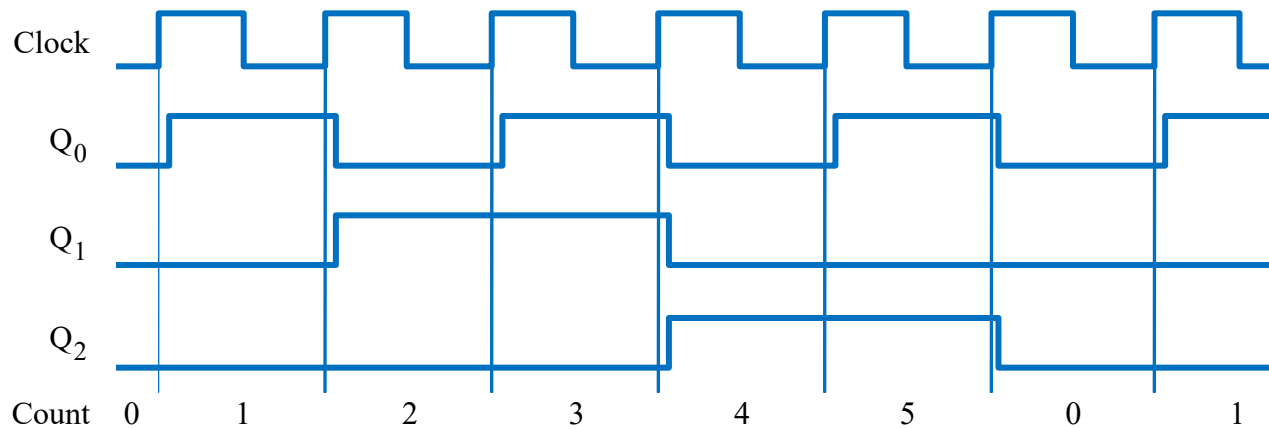


A counter with parallel-load capability

Modulo- n counter



Block diagram

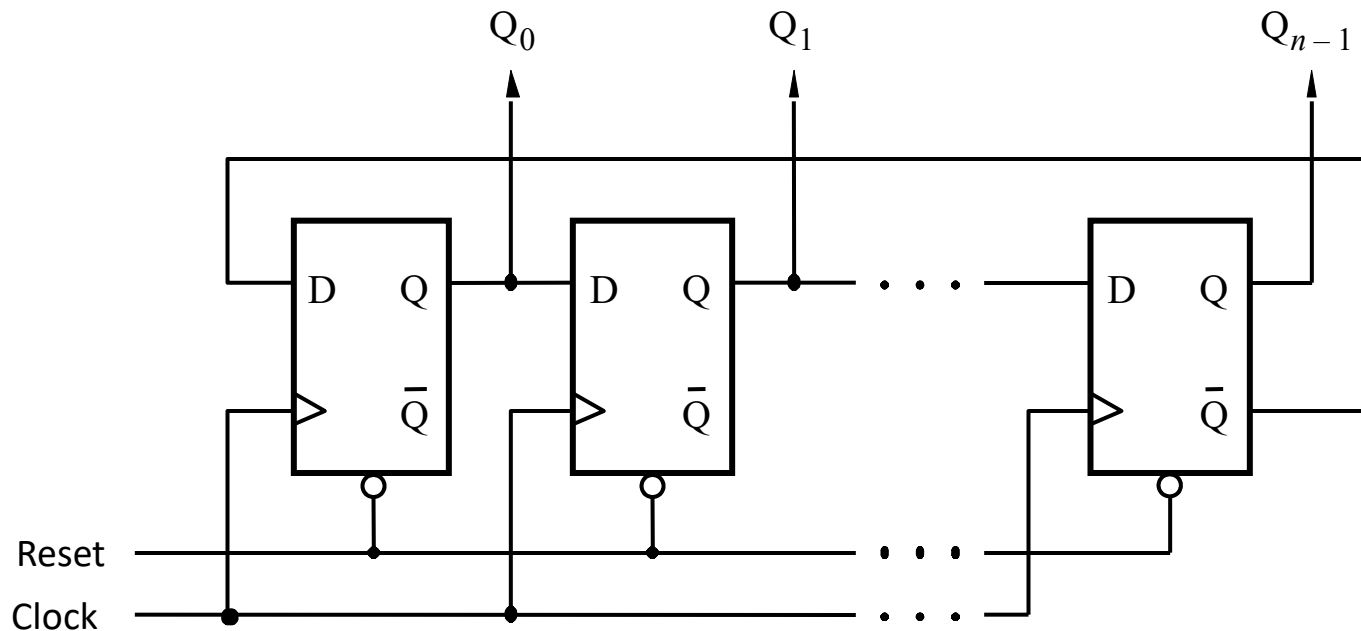


Timing diagram

A modulo-6 counter with synchronous reset.

Johnson Counter

- A 4-bit Johnson counter is shown below.
- It produces the sequence 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, 0000, and so on



Other useful counters

- Ring counter
 - Only one output bit is one at any time
 - A 4-bit ring counter produces the sequence 1000, 0100, 0010, 0001, 1000 and so on
 - It can be designed with a DFF based shift register (shown earlier) with the following two minor changes
 - to start the counter operation, the **first** DFF is Preset to 1 and **all other** DFFs are Cleared to 0 (all asynchronously)
 - the output of the last DFF is connected to the input of the first DFF
- BCD counter (Reading assignment)